# T0 Model: Granulation, Limits and Fundamental Asymmetry

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## Abstract

The T0 model describes a fundamental granulation of spacetime at the sub-Planck scale $\ell_0 = \xi \times \ell_P$ with $\xi \approx 1.333 \times 10^{-4}$. This work examines the consequences for scale hierarchies, time continuity, and the mathematical completeness of various gravitational theories. The time-mass duality $T(x,t) \cdot m(x,t) = 1$ requires both fields to be coupled and variable, while the fundamental $\xi$-asymmetry enables all developmental processes.

## .1 Granulation as Fundamental Principle of Reality

### Minimum Length Scale $\ell_0$

The T0 model introduces a fundamental length scale deeper than the Planck length:

$$\ell_0 = \xi \times \ell_P \approx \frac{4}{3} \times 10^{-4} \times 1.616 \times 10^{-35} \text{ m} \approx 2.155 \times 10^{-39} \text{ m} \tag{1}$$

**Significance of $\ell_0$:**

- Absolute physical lower limit for spatial structures
- Granulated spacetime structure - not continuous
- Sub-Planck physics with new fundamental laws
- Universal scale for all physical phenomena

### The Extreme Scale Hierarchy

From $\ell_0$ to cosmological scales extends a hierarchy of over 60 orders of magnitude:

$$\ell_0 \approx 10^{-39} \text{ m} \quad \text{(Sub-Planck minimum)} \tag{2}$$
$$\ell_P \approx 10^{-35} \text{ m} \quad \text{(Planck length)} \tag{3}$$
$$L_{\text{Casimir}} \approx 100 \text{ micrometers} \quad \text{(Casimir scale)} \tag{4}$$
$$L_{\text{Atom}} \approx 10^{-10} \text{ m} \quad \text{(Atomic scale)} \tag{5}$$
$$L_{\text{Macro}} \approx 1 \text{ m} \quad \text{(Human scale)} \tag{6}$$
$$L_{\text{Cosmo}} \approx 10^{26} \text{ m} \quad \text{(Cosmological scale)} \tag{7}$$

### Casimir Scale as Evidence of Granulation

At the Casimir characteristic scale, first measurable effects appear:

$$L_\xi = \left(\frac{\xi \hbar c}{\rho_{\text{CMB}}}\right)^{1/4} \approx 100 \text{ micrometers} \tag{8}$$

The scale $L_\xi$ is fixed via $\rho_{\text{CMB}} = \xi\hbar c/L_\xi^4$; with $\rho_{\text{CMB}} = 4.175 \times 10^{-14}$ J/m$^3$ this gives $L_\xi = 100.2$ $\mu$m.

**Experimental evidence**:

- Deviations from $1/d^4$ law at distances $\approx 10$ nm
- $\xi$-corrections in Casimir force measurements
- Limits of continuum physics become visible

## .2 Limit Systems and Scale Hierarchies

### Three-Scale Hierarchy

The T0 model organizes all physical scales into three fundamental domains:

1. $\ell_0$**-domain**: Granulated physics, universal laws
2. **Planck domain**: Quantum gravity, transition dynamics
3. **Macro domain**: Classical physics with $\xi$-corrections

### Relational Number System

Prime number ratios organize particles into natural generations:

- **3-limit**: u-, d-quarks (1st generation)
- **5-limit**: c-, s-quarks (2nd generation)
- **7-limit**: t-, b-quarks (3rd generation)

The next prime number (11) leads to $\xi^{11}$-corrections $\approx 2.4 \times 10^{-43}$, which lie below the Planck scale.

### CP Violation from Universal Asymmetry

The $\xi$-asymmetry explains:

- CP violation in weak interactions

- Matter-antimatter asymmetry in the universe
- Chiral symmetry breaking in nature

## .3 Fundamental Asymmetry as Motion Principle

### The Universal $\xi$-Constant

$$\xi = \frac{4}{3} \times 10^{-4} \approx 1.333 \times 10^{-4} \tag{9}$$

**Origin**: Geometric 4/3-constant from optimal 3D space packing
**Effect**: Universal asymmetry enabling all development

### Eternal Universe Without Big Bang

The T0 model describes an eternal, infinite, non-expanding universe:
- No beginning, no end - timeless existence
- Heisenberg's uncertainty principle forbids a classical Big Bang singularity with infinite density: $\Delta E \times \Delta t \geq \hbar/2$; instead, a tiny but finite core with minimal length scale $L_0$ (from $\xi$) remains.
- Structured development instead of chaotic explosion
- Continuous $\xi$-field dynamics instead of Big Bang

### Time Exists Only After Field-Asymmetry Excitation

**Hierarchy of time emergence**:
1. **Timeless universe**: Perfect symmetry, no time
2. **$\xi$-asymmetry arises**: Symmetry breaking activates time field
3. **Time-energy duality**: $T(x,t) \cdot E(x,t) = 1$ becomes active
4. **Manifested time**: Local time emerges through field dynamics
5. **Directed time**: Thermodynamic arrow of time stabilizes

Time is not fundamental but emergent from field asymmetry.

## .4 Hierarchical Structure: Universe > Field > Space

### The Fundamental Order Hierarchy

**Universe (highest order level)**:
- Superordinate structure with eternal, infinite properties
- Global organizational principles determine everything below
- $\xi$-asymmetry as universal guiding structure
- Thermodynamic overall balance of all processes

**Field (middle organizational level)**:
- Universal $\xi$-field as mediator between universe and space
- Local dynamics within global constraints
- Time-energy duality as field principle

- Structure-forming processes through asymmetry

**Space (manifestation level)**:

- 3D geometry as stage for field manifestations
- Granulation at $\ell_0$-scale
- Local interactions between field excitations

### Causal Downward Coupling

$$\text{UNIVERSE} \rightarrow \text{FIELD} \rightarrow \text{SPACE} \rightarrow \text{PARTICLES} \tag{10}$$

The universe is not just the sum of its spatial parts. Superordinate properties emerge only at the highest level. The $\xi$-constant is universal, not a space property.

## .5 Continuous Time Beyond Certain Scales

### The Crucial Scale Hierarchy of Time

In the T0 model, different time domains exist with fundamentally different properties. The further we move from $\ell_0$, the more continuous and constant time becomes.

#### Granulated Zone (below $\ell_0$)

$$\ell_0 = \xi \times \ell_P \approx 2.155 \times 10^{-39} \text{ m} \tag{11}$$

- Time is discretely granulated, not continuous
- Chaotic quantum fluctuations dominate
- Physics loses classical meaning
- All fundamental forces equally strong

#### Transition Zone (around $\ell_0$)

- Time-mass duality $T \cdot m = 1$ becomes fully active
- Intensive interaction of all fields
- Transition from granulated to continuous

#### Continuous Zone (above $\ell_0$)

**Central Insight**

$$\text{Distance to } \ell_0 \uparrow \quad \Rightarrow \quad \text{Time continuity} \uparrow \quad \Rightarrow \quad \text{Constant direction} \uparrow \tag{12}$$

- Beyond a certain point, time becomes continuous
- Constant directed flow direction emerges
- The greater the distance to $\ell_0$, the more stable the time direction
- Emergent classical physics with $\xi$-corrections

### Quantitative Scaling of Time Continuity

**Time continuity as function of distance to $\ell_0$:**

$$\text{Time continuity} \propto \log\left(\frac{L}{\ell_0}\right) \quad \text{for } L \gg \ell_0 \tag{13}$$

**Practical scales**:

$$L = 10^{-35} \text{ m (Planck)}: \quad \text{Still granulated} \tag{14}$$
$$L = 10^{-15} \text{ m (Nuclear)}: \quad \text{Transition to continuity} \tag{15}$$
$$L = 10^{-10} \text{ m (Atomic)}: \quad \text{Practically continuous} \tag{16}$$
$$L = 10^{-3} \text{ m (mm)}: \quad \text{Completely continuous, constant direction} \tag{17}$$
$$L = 1 \text{ m (Meter)}: \quad \text{Perfectly linear, directed time} \tag{18}$$

### Thermodynamic Arrow of Time

**Scale-dependent entropy**:
- **Granulated level ($\ell_0$)**: Maximum entropy, perfect symmetry
- **Transition level**: Entropy gradients emerge
- **Continuous level**: Second law becomes active
- **Macroscopic level**: Irreversible time direction

## .6 Practical vs. Fundamental Physics

### Time is Practically Experienced as Constant

De facto for us: Time flows constantly in our experience domain
- **Local scales (m to km)**: Time is practically perfectly linear and constant
- **Measurable variations**: Only under extreme conditions (GPS satellites, particle accelerators)
- **Everyday physics**: Time constancy is a good approximation

### Speed of Light as Clear Upper Limit

**Observed reality**:
- $c = 299,792,458$ m/s is measurable upper limit for information transfer
- **Causality**: No signals faster than $c$ observed
- **Relativistic effects**: Clearly measurable at $v \to c$
- **Particle accelerators**: Confirm $c$-limit daily

### Resolution of the Apparent Contradiction

**Macroscopic level (our world)**:

$$L = 1 \text{ m to } 10^6 \text{ m (km range)} \tag{19}$$

- Time flows constantly: $dt/dt_0 \approx 1 + 10^{-16}$ (very small, but measurable with optical clocks)

- $c$ is practically constant: $\Delta c/c \approx 10^{-16}$ (very small)
- Einstein physics works perfectly

**Fundamental level (T0 model)**:

$$\ell_0 = 10^{-39} \text{ m to } \ell_P = 10^{-35} \text{ m} \tag{20}$$

- Time-mass duality: $T \cdot m = 1$ is fundamental
- $c$ is ratio: $c = L/T$ (must be variable)
- Mathematical consistency requires coupled variation

**These variations are very small. What optical clocks resolve today – frequency comparisons at $10^{-18}$, the gravitational redshift over 33 cm ($\approx 4 \times 10^{-17}$, Chou et al. 2010) – is the gravitational redshift of General Relativity. A signature of the variation beyond it has not been derived so far (Doc. 129, 392); this does not rule it out.**

A variation of $c$ is to be read here as *fractal path lengthening*: as a geometric stretching of the propagation path in the $\xi$ field (Doc. 026, 149, 182), which is achromatic, not as a change of $c$ itself.

## .7 Gravitation: Mass Variation vs. Space Curvature

### Two Equivalent Interpretations

**Einstein interpretation**:

- $m$ = constant (fixed mass)
- $g_{\mu\nu}$ = variable (curved spacetime)
- Mass causes space curvature

**T0 interpretation**:

- $m(x,t)$ = variable (dynamic mass)
- $g_{\mu\nu}$ = fixed (flat Euclidean space)
- Mass varies locally through $\xi$-field

### Important Insight: We Don't Know!

**Attention - Fundamental Point**

We DO NOT KNOW whether mass causes space curvature or whether mass itself varies!
This is an assumption, not a proven fact!

**Both interpretations are equally valid**:
**Einstein assumption**:

$$\text{Mass/energy} \rightarrow \text{Space curvature} \rightarrow \text{Gravitation} \tag{21}$$

$$G_{\mu\nu} = 8\pi T_{\mu\nu} \tag{22}$$

**T0 alternative**:

$$\xi\text{-field} \rightarrow \text{Mass variation} \rightarrow \text{Gravitational effects} \tag{23}$$

$$m(x,t) = m_0 \cdot (1 + \xi \cdot \Phi(x,t)) \tag{24}$$

Here and in the pragmatic interpretation $m(x) = m_0(1 + \xi \cdot \text{Gravitational field}(x))$ below, $\Phi$ is taken with the opposite sign to the section on mathematical completeness: there $T = T_0(1+\xi\Phi)$ goes with $m = m_0(1 - \xi\Phi)$, as $T \cdot m = 1$ requires.

### Experimental Indistinguishability

**All measurements are frequency-based**:

- **Clocks**: Hyperfine transition frequencies
- **Scales**: Spring oscillations/resonance frequencies
- **Spectrometers**: Light frequencies and transitions
- **Interferometers**: Phases = frequency integrals

**Identical frequency shifts**:

$$\text{Einstein}: \quad \nu' = \nu_0\sqrt{1 + 2\Phi/c^2} \approx \nu_0(1 + \Phi/c^2) \tag{25}$$

$$\text{T0}: \quad \nu' = \nu_0 \cdot \frac{m(x,t)}{T(x,t)} \approx \nu_0(1 + \Phi/c^2) \tag{26}$$

Only frequency ratios are measurable - absolute frequencies are fundamentally inaccessible!

## .8 Mathematical Completeness: Both Fields Coupled Variable

### The Correct Mathematical Formulation

**Mathematically correct in T0 model**:

$$T(x,t) = \text{variable} \quad \text{(Time as dynamic field)} \tag{27}$$

$$m(x,t) = \text{variable} \quad \text{(Mass as dynamic field)} \tag{28}$$

**Coupled through fundamental duality**:

$$T(x,t) \cdot m(x,t) = 1 \tag{29}$$

**Both fields vary TOGETHER**:

$$T(x,t) = T_0 \cdot (1 + \xi \cdot \Phi(x,t)) \tag{30}$$

$$m(x,t) = m_0 \cdot (1 - \xi \cdot \Phi(x,t)) \tag{31}$$

### Verification of Mathematical Consistency

**Duality check**:

$$T(x,t) \cdot m(x,t) = T_0 m_0 \cdot (1 + \xi\Phi)(1 - \xi\Phi) \tag{32}$$

$$= T_0 m_0 \cdot (1 - \xi^2\Phi^2) \tag{33}$$

$$\approx T_0 m_0 = 1 \quad (\text{for } \xi\Phi \ll 1) \tag{34}$$

Mathematical consistency confirmed!

### Why Both Fields Must Be Variable

**Lagrange formalism requires**:

$$\delta S = \int \delta \mathscr{L} d^4x = 0 \tag{35}$$

**Complete variation**:

$$\delta \mathscr{L} = \frac{\partial \mathscr{L}}{\partial T}\delta T + \frac{\partial \mathscr{L}}{\partial m}\delta m + \frac{\partial \mathscr{L}}{\partial \partial_\mu T}\delta \partial_\mu T + \frac{\partial \mathscr{L}}{\partial \partial_\mu m}\delta \partial_\mu m \tag{36}$$

For mathematical completeness:

- $\delta T \neq 0$ (Time must be variable)
- $\delta m \neq 0$ (Mass must be variable)
- Both coupled through $T \cdot m = 1$

### Einstein's Arbitrary Constant Setting

Einstein arbitrarily sets:

$$m_0 = \text{constant} \quad \Rightarrow \quad \delta m = 0 \tag{37}$$

**Mathematical problem**:

- Incomplete variation of the Lagrangian
- Violates variation principle of field theory
- Arbitrary symmetry breaking without justification

### Parameter Elegance

$$\text{Einstein}: \quad m_0, c, G, \hbar, \Lambda, \alpha_{\text{EM}}, \ldots \quad (\gg 10 \text{ free parameters}) \tag{38}$$

$$\text{T0}: \quad \xi \quad (1 \text{ universal parameter}) \tag{39}$$

## .9 Pragmatic Preference: Variable Mass with Constant Time

### The Pragmatic Alternative for Our Experience Space

As pragmatists, one can certainly prefer:

$$\text{Time}: \quad t = \text{constant} \quad \text{(practical experience)} \tag{40}$$

$$\text{Mass}: \quad m(x,t) = \text{variable} \quad \text{(dynamic adjustment)} \tag{41}$$

**Why this is pragmatically sensible**:

- Time constancy corresponds to our direct experience
- Mass variation is conceptually easier to imagine
- Practical calculations often become simpler
- Intuitive understandability for applications

### Practical Advantages of Constant Time

In our experienceable space (m to km):

- Time flows linearly and constantly - our direct experience
- Clocks tick uniformly - practical time measurement
- Causal sequences are clearly defined
- Technical applications (GPS, navigation) function

**Language convention**:

- Time passes constantly
- Mass adapts to the fields
- Matter becomes heavier/lighter depending on location

### Variable Mass as Intuitive Concept

**Pragmatic interpretation**:

$$m(x) = m_0 \cdot (1 + \xi \cdot \text{Gravitational field}(x)) \tag{42}$$

**Intuitive conception**:

- Mass increases in strong gravitational fields
- Mass decreases in weaker fields
- Matter feels the local $\xi$-field
- Dynamic adaptation to environment

### Scientific Legitimacy of Preference

**Important Insight**

Pragmatic preferences are scientifically justified when both approaches are experimentally equivalent!

**Justification**:

- Scientifically equivalent to Einstein approach
- Often practically advantageous for applications
- Didactically easier to teach
- Technically more efficient to implement

The choice between constant time + variable mass vs. Einstein is a matter of taste - both are scientifically equally justified!

## .10 The Eternal Philosophical Boundary

### What the T0 Model Explains

- HOW the $\xi$-asymmetry works
- WHAT the consequences are
- WHICH laws follow from it
- WHEN time and development emerge

### What the T0 Model CANNOT Explain

The fundamental questions remain:
- WHY does the $\xi$-asymmetry exist?
- WHERE does the original energy come from?
- WHO/WHAT gave the first impulse?
- WHY does anything exist at all instead of nothing?

### Scientific Humility

**The eternal boundary**: Every explanation needs unexplained axioms. The ultimate reason always remains mysterious. The that of existence is given, the why remains open.

**The elegant shift**: The T0 model shifts the mystery to a deeper, more elegant level - but it cannot resolve the fundamental riddle of existence.

And that is good. Because a universe without mystery would be a boring universe.

## .11 Experimental Predictions and Tests

### Casimir Effect Modifications

- Deviations from $1/d^4$ law at $d \approx 10$ nm
- $\xi$-corrections in precision measurements
- Frequency-dependent Casimir forces

### Atom Interferometry

- $\xi$-resonances in quantum interferometers
- Mass variations in gravitational fields
- Time-mass duality in precision experiments

### Gravitational Wave Detection

- $\xi$-corrections in LIGO/Virgo data
- Modifications of wave dispersion
- Sub-Planck structures in gravitational waves

## .12 Mathematical Proof: The Formula $T \cdot m = 1$ Excludes Singularities

### Important Clarification: $T$ as Oscillation Period

**ATTENTION:** In this analysis, $T$ does not mean the experienced, continuously flowing time, but the **oscillation period** or **characteristic time constant** of a system. This is a fundamental difference:
- $T =$ oscillation period (discrete, characteristic time unit)
- Not: $T =$ continuous time coordinate (our everyday experience)

### The Fundamental Exclusion Property

The equation $T \cdot m = 1$ is not just a mathematical relationship – it is an **exclusion theorem**. Through its algebraic structure, it makes certain states mathematically impossible.

### Proof 1: Exclusion of Infinite Mass

**Assumption:** There exists an infinite mass $m = \infty$
**Mathematical consequence:**

$$T \cdot m = 1 \tag{43}$$
$$T \cdot \infty = 1 \tag{44}$$
$$T = \frac{1}{\infty} = 0 \tag{45}$$

**Contradiction:** $T = 0$ is not in the domain of the equation $T \cdot m = 1$, since:

- The product $0 \cdot \infty$ is mathematically undefined
- The original equation $T \cdot m = 1$ would be violated $(0 \cdot \infty \neq 1)$

**Conclusion:** $m = \infty$ is excluded by the formula.

### Proof 2: Exclusion of Infinite Time

**Assumption:** There exists an infinite time $T = \infty$
**Mathematical consequence:**

$$T \cdot m = 1 \tag{46}$$
$$\infty \cdot m = 1 \tag{47}$$
$$m = \frac{1}{\infty} = 0 \tag{48}$$

**Contradiction:** $m = 0$ is not in the domain, since:

- The product $\infty \cdot 0$ is mathematically undefined
- The equation $T \cdot m = 1$ would be violated $(\infty \cdot 0 \neq 1)$

**Conclusion:** $T = \infty$ is excluded by the formula.

### Proof 3: Exclusion of Zero Values

**Assumption:** There exists $T = 0$ or $m = 0$
**Case 1:** $T = 0$

$$T \cdot m = 1 \Rightarrow 0 \cdot m = 1 \tag{49}$$

This is impossible for any finite value of $m$, since $0 \cdot m = 0 \neq 1$.
**Case 2:** $m = 0$

$$T \cdot m = 1 \Rightarrow T \cdot 0 = 1 \tag{50}$$

This is impossible for any finite value of $T$, since $T \cdot 0 = 0 \neq 1$.
**Conclusion:** Both $T = 0$ and $m = 0$ are excluded by the formula.

## Proof 4: Exclusion of Mathematical Singularities

**Definition of a singularity:** A point where a function becomes undefined or infinite.

**Analysis of the function $T = \frac{1}{m}$:**

**Potential singularities could occur at:**

- $m = 0$ (division by zero)
- $T \to \infty$ (infinite function values)

**Exclusion by the constraint $T \cdot m = 1$:**

1. **At $m = 0$**: The equation $T \cdot m = 1$ cannot be satisfied
2. **At $T \to \infty$**: Would require $m \to 0$, which is already excluded

**Mathematical proof of singularity freedom:**

For every point $(T, m)$ with $T \cdot m = 1$:

$$T = \frac{1}{m} \text{ with } m \in (0, +\infty) \tag{51}$$

$$m = \frac{1}{T} \text{ with } T \in (0, +\infty) \tag{52}$$

Both functions are on their entire domain:

- **Continuous**
- **Differentiable**
- **Finite Well-defined**

## The Algebraic Protection Function

The equation $T \cdot m = 1$ acts like an **algebraic protection** against singularities:

### Automatic Correction

$$\text{If } m \text{ becomes very small} \Rightarrow T \text{ automatically becomes very large} \tag{53}$$

$$\text{If } T \text{ becomes very small} \Rightarrow m \text{ automatically becomes very large} \tag{54}$$

$$\text{But: } T \cdot m \text{ always remains exactly } 1 \tag{55}$$

### Mathematical Stability

$$\lim_{m \to 0^+} T = +\infty, \text{ but } T \cdot m = 1 \text{ remains satisfied} \tag{56}$$

$$\lim_{T \to 0^+} m = +\infty, \text{ but } T \cdot m = 1 \text{ remains satisfied} \tag{57}$$

The constraint **forces** the variables into a finite, well-defined region.

## Proof 5: Positive Definiteness

**Theorem:** All solutions of $T \cdot m = 1$ are positive.

**Proof:**

$$T \cdot m = 1 > 0 \tag{58}$$

Since the product is positive, both factors must have the same sign.

**Exclusion of negative values:**

- If $T < 0$ and $m < 0$, then $T \cdot m > 0$, but physically meaningless
- If $T > 0$ and $m < 0$, then $T \cdot m < 0 \neq 1$
- If $T < 0$ and $m > 0$, then $T \cdot m < 0 \neq 1$

**Conclusion:** Only $T > 0$ and $m > 0$ satisfy the equation.

## The Fundamental Insight About Time and Continuity

**Important physical clarification:**

The formula $T \cdot m = 1$ describes **discrete, characteristic properties** of systems, not the continuous time flow of our experience. This means:

**What $T \cdot m = 1$ does NOT state:**

- „Time stands still" $(T = 0)$
- „Processes take infinitely long" $(T = \infty)$
- „The time flow is interrupted"
- „Our experienced time disappears"

**What $T \cdot m = 1$ actually describes:**

- **Oscillation periods** have mathematical limits
- **Characteristic time constants** cannot become arbitrary
- **Discrete time units** stand in fixed relation to mass
- **Periodic processes** follow the constraint $T \cdot m = 1$

**The continuous time flow remains unaffected**

The continuous time coordinate $t$ (our „arrow time") is **not affected** by this relationship. $T \cdot m = 1$ regulates only the **intrinsic time scales** of physical systems, not the superordinate time flow in which these systems exist.

**Important insight about our time perception:**

Our continuous time perception could practically be only a **tiny excerpt** of a much larger period – an oscillation period so immense that it far exceeds anything humans could ever experience or conceive.

**Conceivable orders of magnitude:**

- **Human life:** $\sim 10^2$ years
- **Human history:** $\sim 10^4$ years
- **Earth age:** $\sim 10^9$ years
- **Universe age:** $\sim 10^{10}$ years **Possible cosmic period:** $10^{50}$, $10^{100}$ or even larger time scales

In such a scenario, our entire observable universe would experience only an **infinitesimal small fraction** of a fundamental oscillation period. For us, time appears linear and continuous because we perceive only a vanishingly small section of a huge cosmic „oscillation".

**Analogy:** Just as a bacterium on a clock hand would perceive the movement as „straight ahead", although it moves on a circular path, we might experience „linear time", although we are in a gigantic periodic structure.

This perspective shows that $T \cdot m = 1$ and our time perception can operate on completely different scales without contradicting each other.

## Cosmological Implications

**This viewpoint opens new possibilities:**

What we observe as cosmic development and change could be only a **small section** in a much larger cyclic pattern that follows the fundamental relationship $T \cdot m = 1$.

**Possible cosmic structure:**

- **Local time perception:** Linear, continuous (our experience domain)
- **Middle time scales:** Observable cosmic developments
- **Fundamental time scale:** Gigantic period according to $T \cdot m = 1$

**Implications:**

- Nature could be organized in **layered-periodic** fashion
- Different time scales follow different regularities
- $T \cdot m = 1$ could be the **master constraint** for the largest scale
- Our observable cosmic development would be a fragment of a cyclic system

This interpretation shows how mathematical constraints $(T \cdot m = 1)$ and physical observations (linear time perception) can coexist in a **hierarchical time model**.